

REVIEW OF MASTER'S THESIS

Master's student: Assanova Sandugash

Educational program: Applied Data Analytics

Thesis title: Development of Machine Learning Models to Improve the Predictive Ability and Accuracy of Forecasting Annual Medical Care Volume

1. Relevance of the Research Topic

The topic of the master's thesis is relevant because accurate forecasting of healthcare expenditure is a genuine planning need for health ministries, particularly in transitional economies where structural reforms and episodic fiscal shocks, such as the COVID-19 pandemic, disrupt historical spending patterns. Kazakhstan presents a demanding test case: a short administrative time series, severe multicollinearity among health system indicators, and significant structural breaks make conventional forecasting approaches unreliable. Investigating whether machine learning methods can improve forecast accuracy under these conditions, and identifying the data conditions under which they do so, is a timely and practically motivated research direction.

2. Scientific Novelty

The scientific novelty lies in the first systematic comparison of classical ARIMA and ensemble machine learning methods for annual healthcare expenditure forecasting in Kazakhstan. The study introduces a two-track evaluation design that applied separately to the national time series and to the regional administrative panel. It explicitly documents the data-volume threshold at which ML methods become competitive. The use of the regional panel as the primary ML training corpus, with the national series as the forecast target, is a methodologically sound and transferable strategy for other data-scarce health systems.

3. Structure and Content of the Thesis

The introductory chapter defines the research problem, hypotheses, and scope. The literature review covers classical time-series methods, ensemble ML, temporal validation, and the Kazakhstan healthcare planning context, and identifies the specific gaps the study addresses. The methodology chapter describes the data sources, preprocessing steps, exploratory analysis, feature engineering for both pipelines, and the full evaluation protocol. The results chapter presents performance metrics across both tracks, an ablation study, feature importance analysis, multicollinearity diagnostics, inflation sensitivity, and three-year forecasts with uncertainty intervals. The discussion and conclusion chapters interpret the findings in relation to the stated hypotheses and translate them into planning-relevant recommendations.

4. Strengths of the Work

The two-track evaluation design is the most valuable aspect of the work. By maintaining parallel experiments at the regional and national levels, the study produces a clear and honest account of when ML methods outperform classical approaches and when they do not: a distinction that single-track studies routinely obscure. The ablation study effectively isolates the contribution of the autoregressive lag term from the structural health system indicators, providing a mechanistic explanation for the observed performance patterns. The introduction of a walk-forward evaluation protocol for ARIMA addresses a genuine methodological asymmetry and strengthens the basis for comparing ML and classical methods on equal terms. The multicollinearity analysis and its explicit implications for feature attribution reflect a level of statistical discipline that goes beyond standard applied ML practice. The inflation sensitivity analysis adds practical relevance by separating nominal from real expenditure growth.

5. Comments and Suggestions

For patterns gathered from training regions to transfer to the held-out region, the panel evaluation makes the assumption that there is enough homogeneity across regions. This assumption seems to be weakest for administratively distinctive regions, and the methodological section could help with a longer explanation of the situations in which geographic transferability would not work. The three-year projections are based on linear feature extrapolation, which ignores macroeconomic shocks and policy discontinuities. The forecast section would be more useful if it included a quick scenario analysis utilizing several trajectories.

6. Conclusion

Overall, the master's thesis is completed at a high level and deserves the grade «A». The work addresses a well-defined research gap with a methodologically sound design, makes an original contribution to the healthcare forecasting literature in Kazakhstan, and produces results that are both statistically interpretable and practically applicable.

Reviewer:

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